

Verification of whether the final matrix is in rref

1. Row 1, Row 2 and Row 3 are non-zero rows. Row 4 is a zero row.
∴ The condition that all non-zero rows must be above the zero rows is

True.

2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 1 in Column 3
Row 3 Leading Entry = 1 in Column 5

∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is

True.

3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
Row 3 Column 3 = Row 4 Column 3 = 0
Row 4 Column 5 = 0

∴ The condition that for each column having a leading entry, all positions below the leading entry must contain 0 is True.

4. The condition that all leading entries must be 1 is True.

5. Row 1 Column 5 = Row 2 Column 5 = 0
Row 1 Column 3 = 0

∴ The condition that for each column having a leading entry, all positions above the leading entry must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & -4 & 0 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ is in rref.

By defn, Columns 2, 4 and 6 have no pivots. ∴ By defn, x_2 , x_4 and x_6 are free variables.

The corresponding system of equations are: $x_1 - 4x_2 + 5x_6 = 0$, $x_3 - x_6 = 0$ and $x_5 - 4x_6 = 0$. ∴ The solution set is: $x_1 = 4x_2 - 5x_6$, $x_3 = x_6$ and $x_5 = 4x_6$, x_4 is a free variable, x_6 is a free variable.